

E-Beam and RIT Chip Data Analysis

Presented By: Vandana Arora

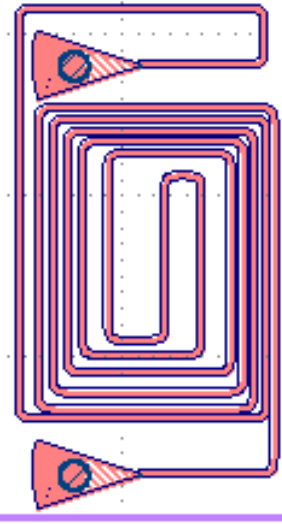
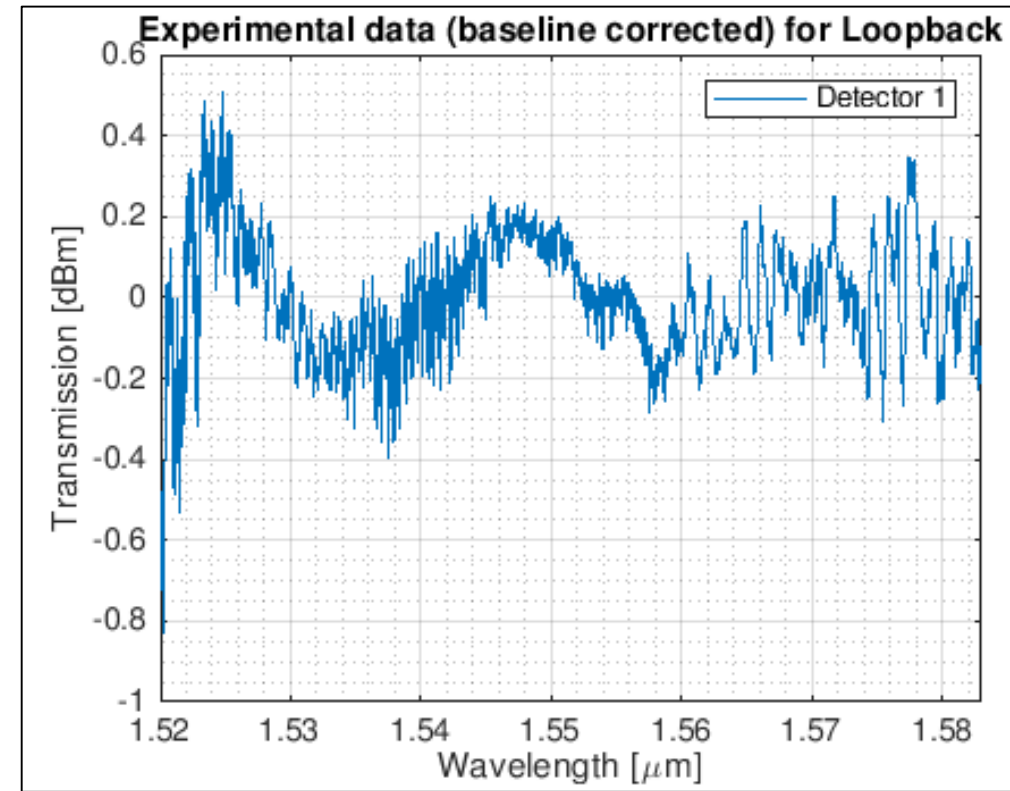
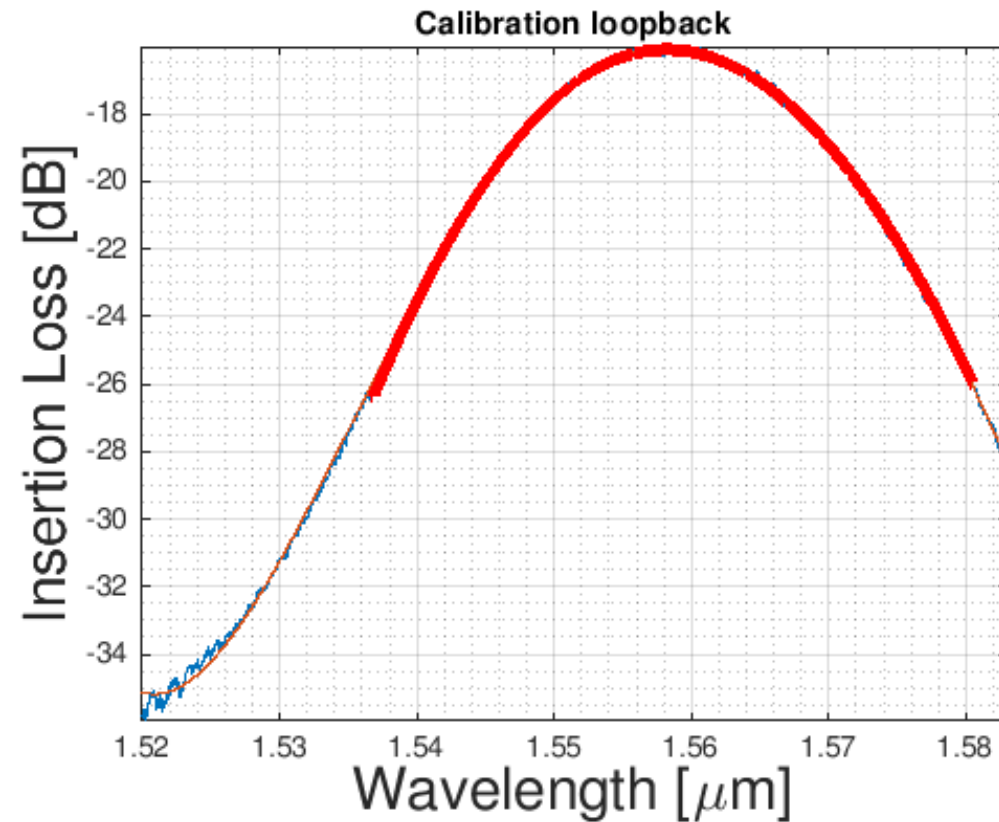
Photonic Integrated Circuits

12/06/2018

Outline

1. **E-Beam Data**
2. RIT Chip Data

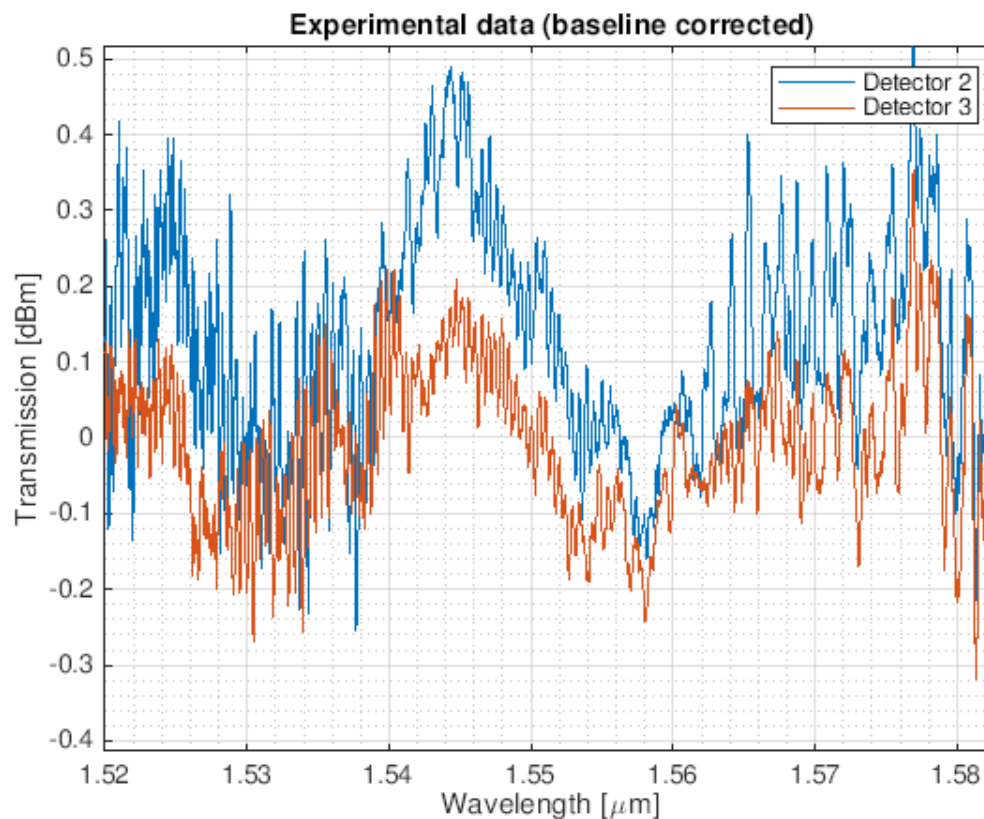
Polarization: TE

Loopback Length: 1784.73 μm Bend Radius: 5 μm 

Minimum loss observed at 1.56 μm $\approx$ 17 dB
Loss increases at very low or very high wavelengths.

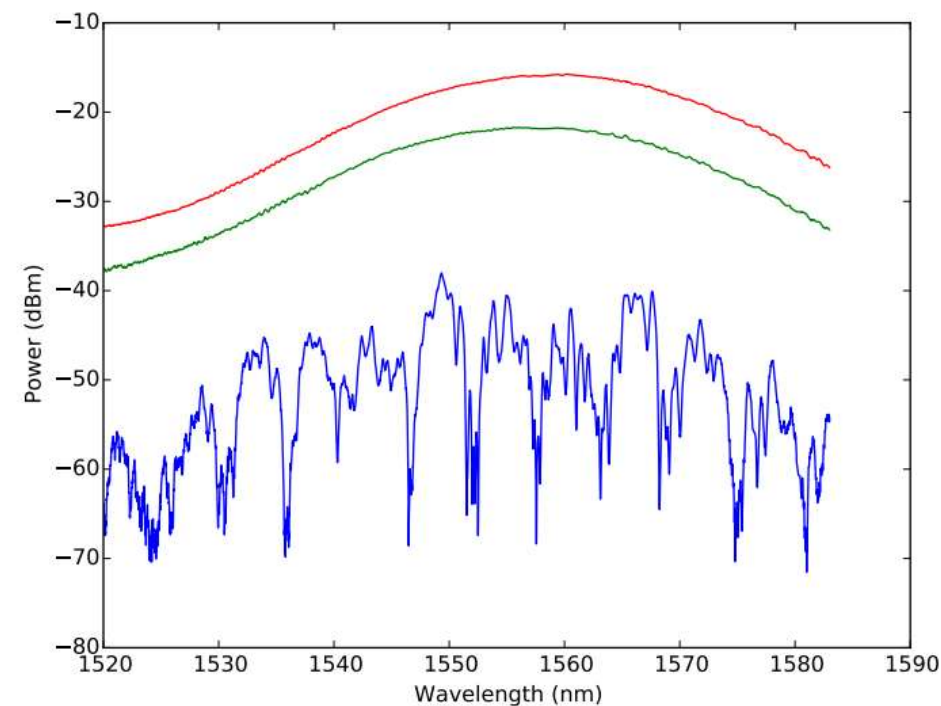
Length Difference $\Delta L = 0 \mu$

Polarization: TE



MZI is balanced but slightly unbalanced. Ideally, no power is expected in detector 2.

Detector 3: Expecting all the power

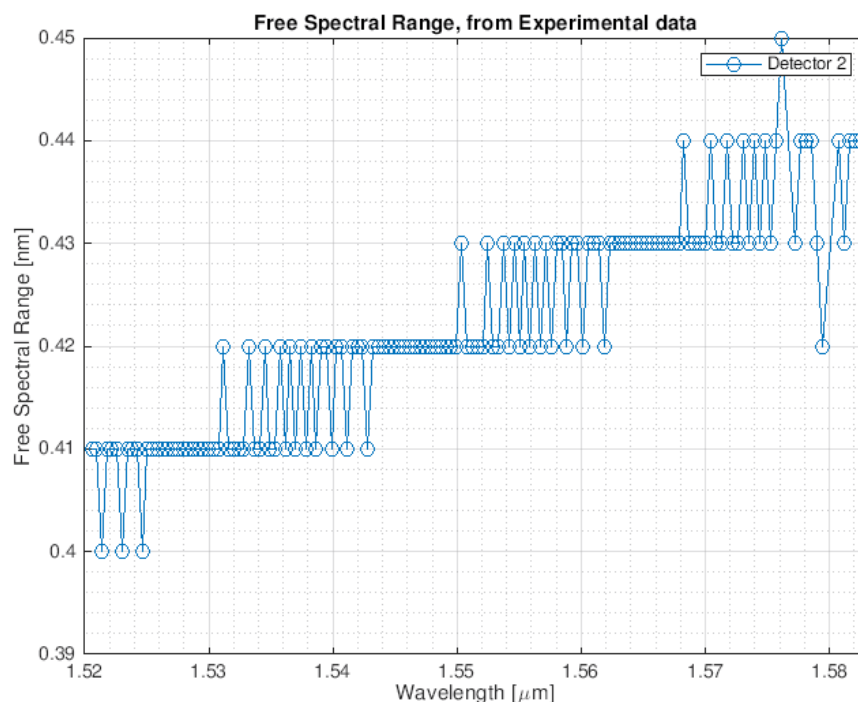
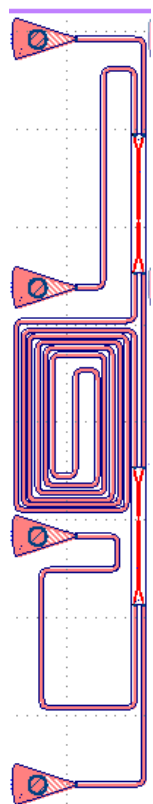


E-Beam Raw Data with Grating Coupler Response

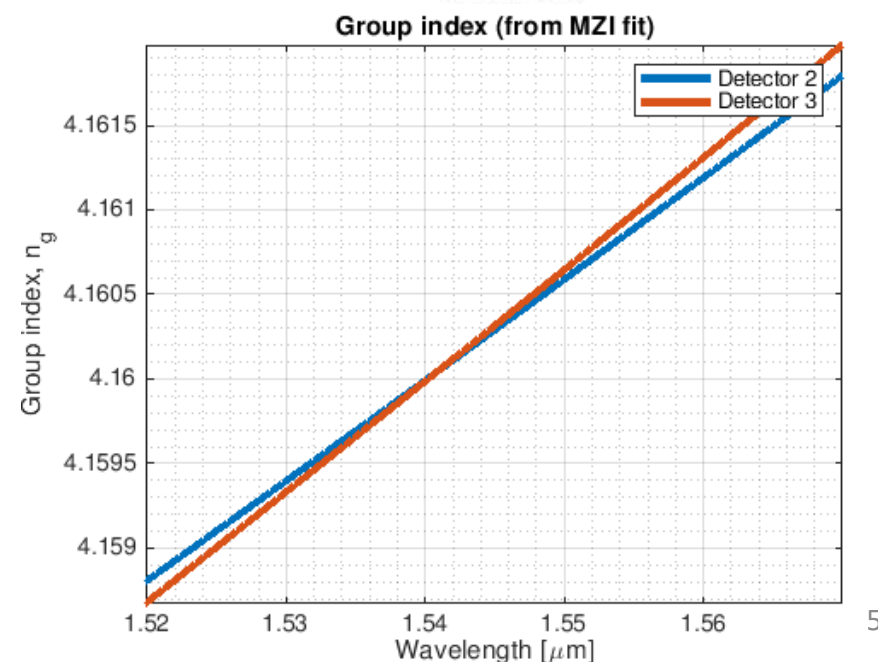
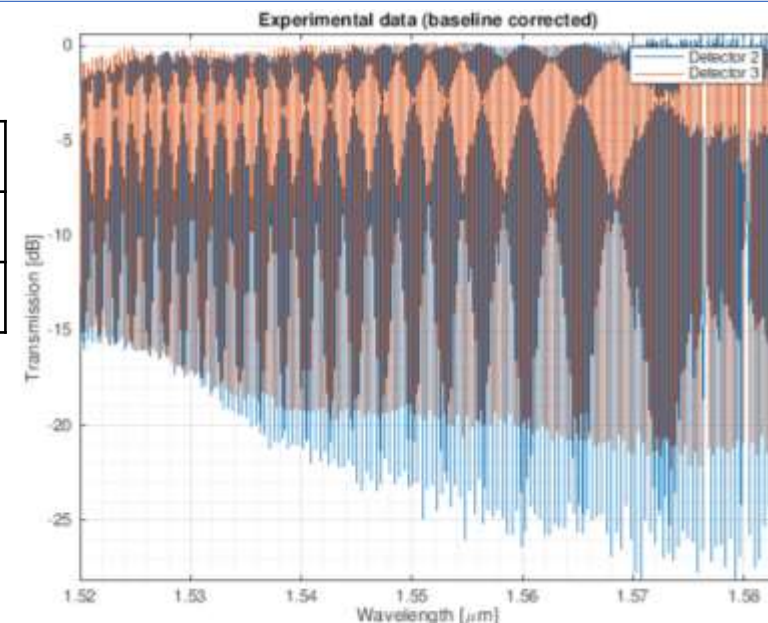
Unbalanced MZI Using Two 50:50 Broadband Directional Couplers

Length Difference $\Delta L = 1369.15 \mu\text{m}$ Polarization: TE

Detector 2	Detector 3
Fit parameters (n1, n2, n3): (2.6, -1, 0)	Fit parameters (n1, n2, n3): (2.65, -1, 0)
$n_g = 4.1603 @ \lambda = 1.545 \mu\text{m}$	$n_g = 4.1603 @ \lambda = 1.545 \mu\text{m}$



$\text{FSR} \approx 0.42 \text{ nm} @ \lambda = 1550 \text{ nm}$
 Dispersion [ps/nm-km]: 673.8243



Unbalanced MZI With Four Rings

MZI Length Difference $\Delta L = 288.603 \mu$

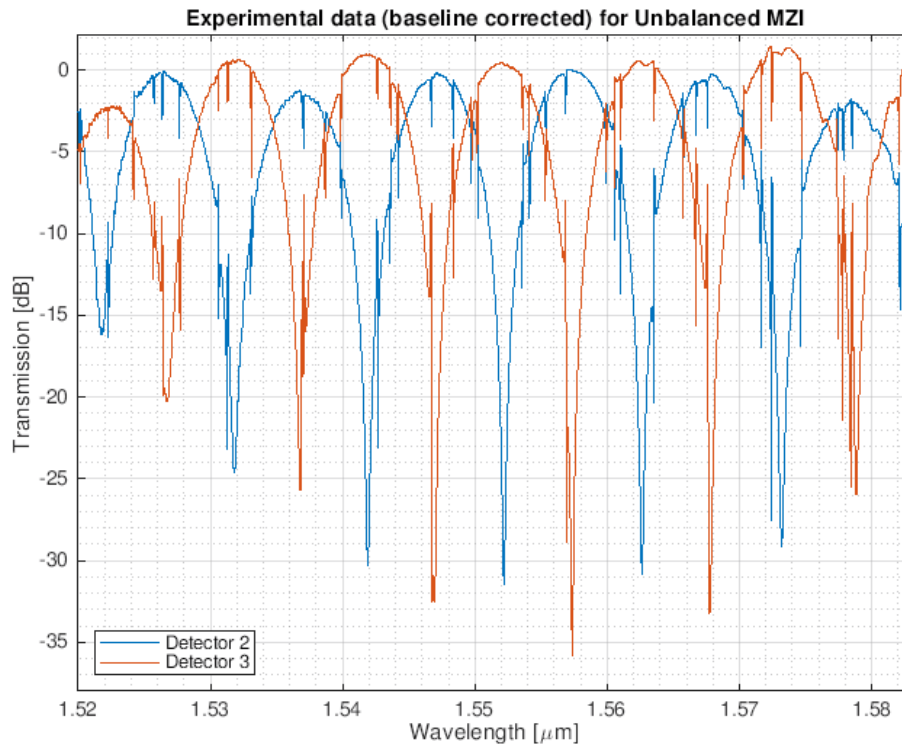
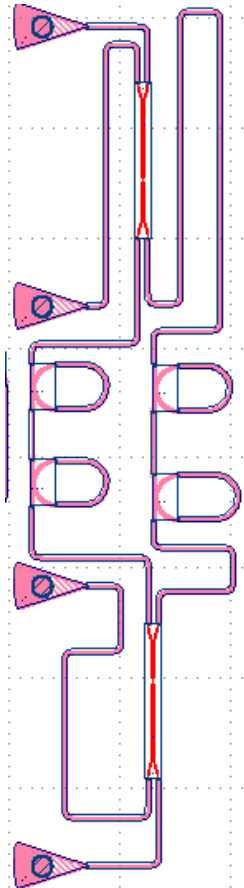
Rings Gap: 200 nm Coupling coefficient $\kappa = 0.04096$

Rings Radius: 10μ Ring Circumference $= 2\pi R = 2\pi * 10 = 62.4 \mu$

From simulation result:

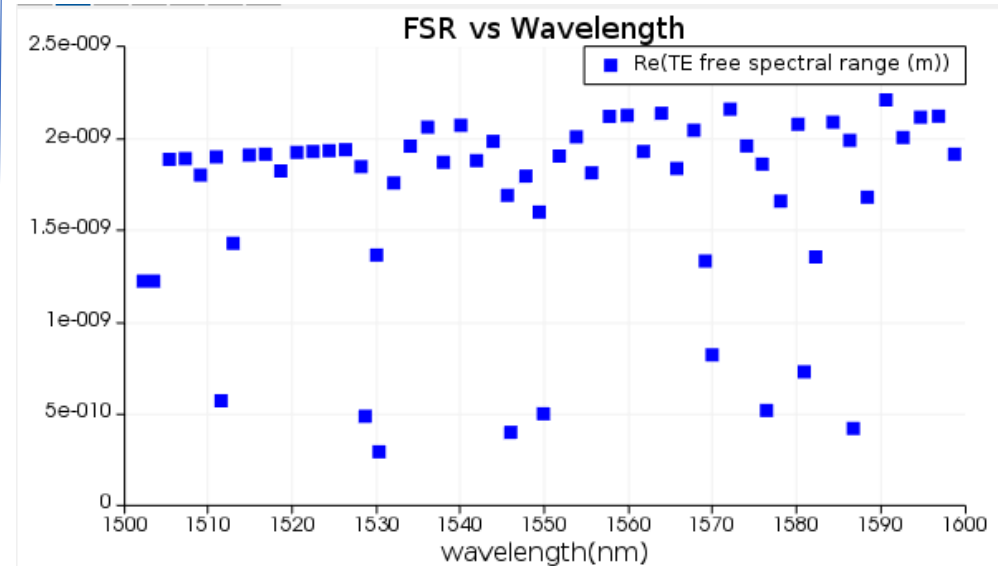
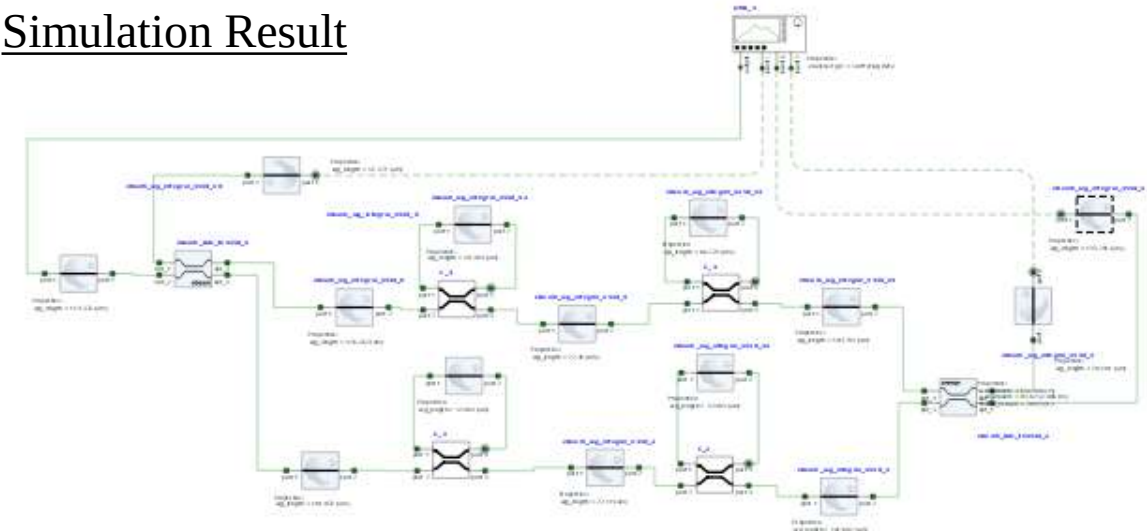
$\text{FSR}@ \lambda = 1.55 \mu\text{m} \approx 2 \text{ nm}$

$n_g = (1.55^2 * 1000) / (2 * 288.6) = 4.16$



Ebeam Data: Group Indices not found using MATLAB script.

Simulation Result



Unbalanced MZIs Using Y-Splitters

MZI1 Length Difference

$$\Delta L_1 = 22.5 \mu$$

MZI2 Length Difference

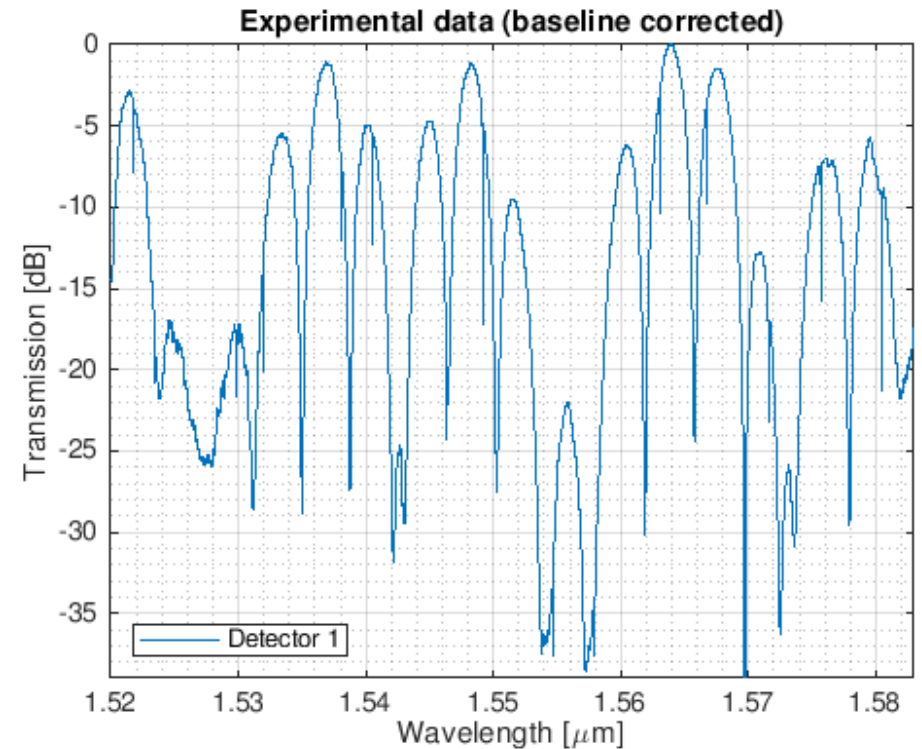
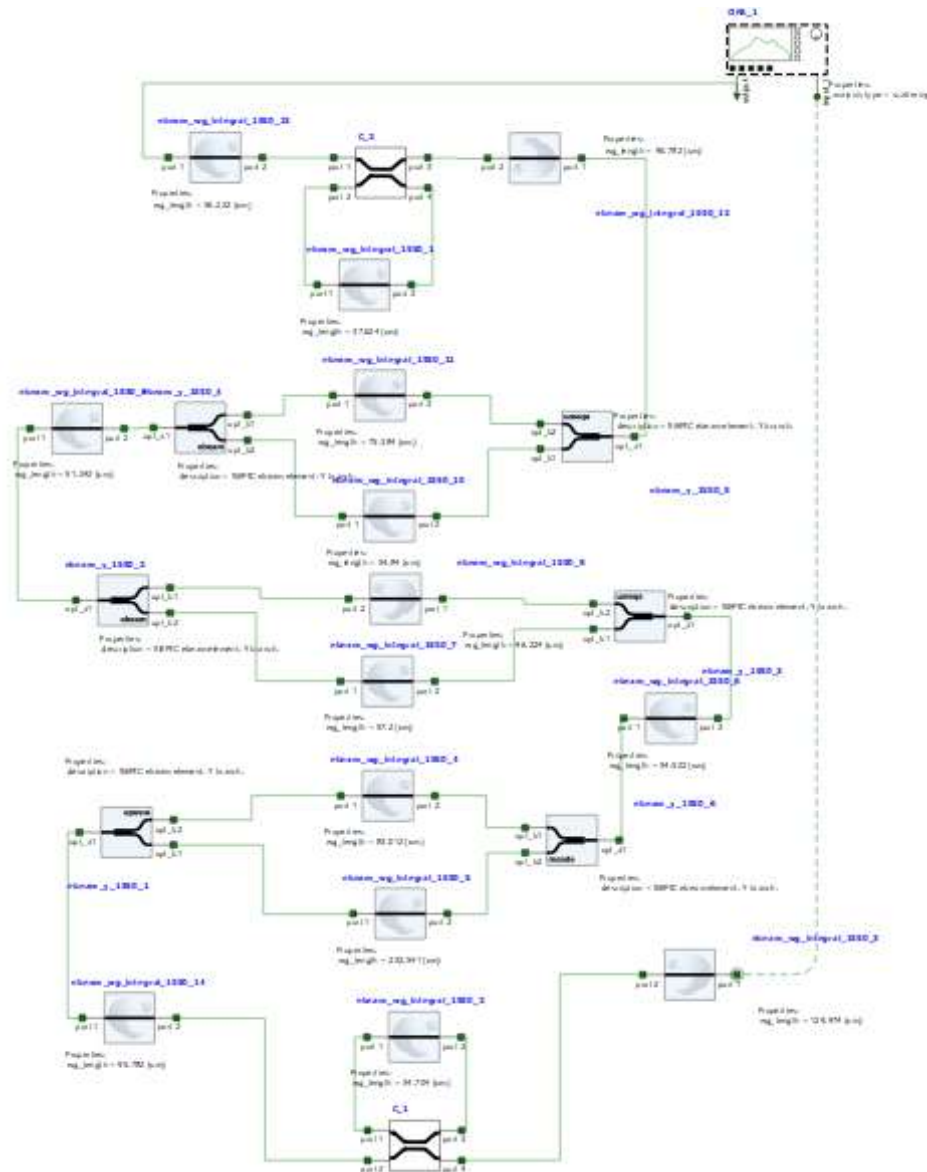
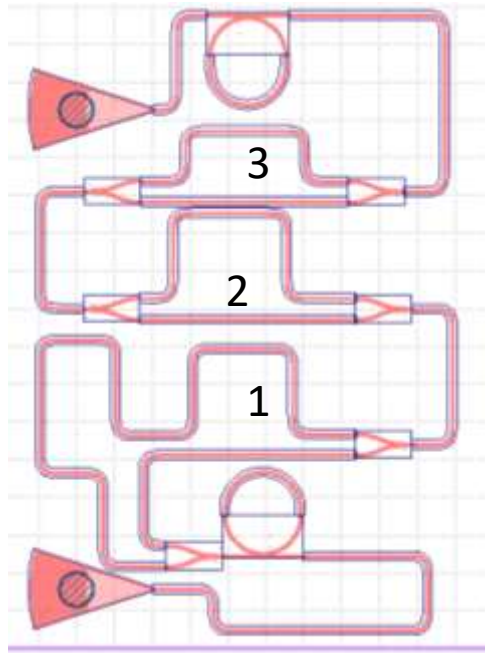
$$\Delta L_2 = 39.024 \mu$$

MZI3 Length Difference

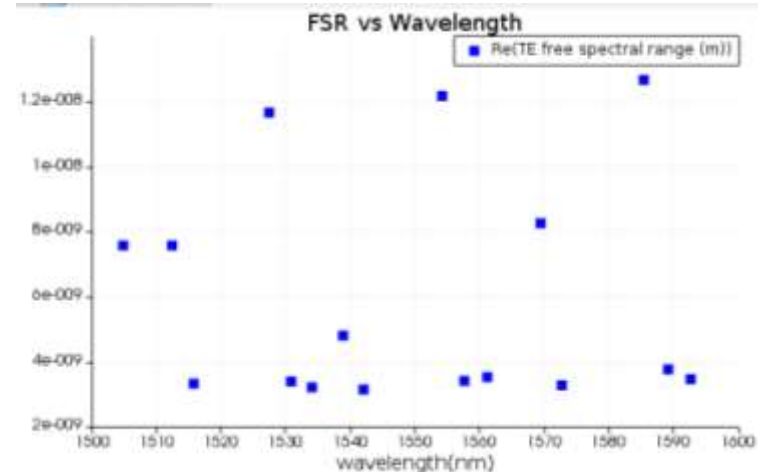
$$\Delta L_3 = 20.54 \mu$$

Rings Gap: 200 nm

Rings Radius: 10 μ



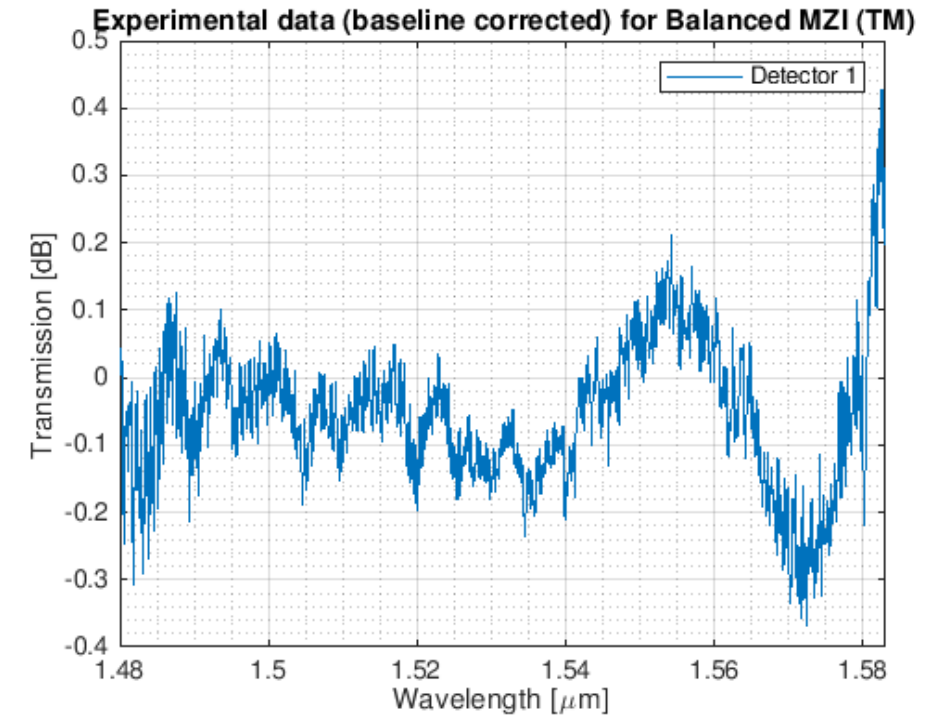
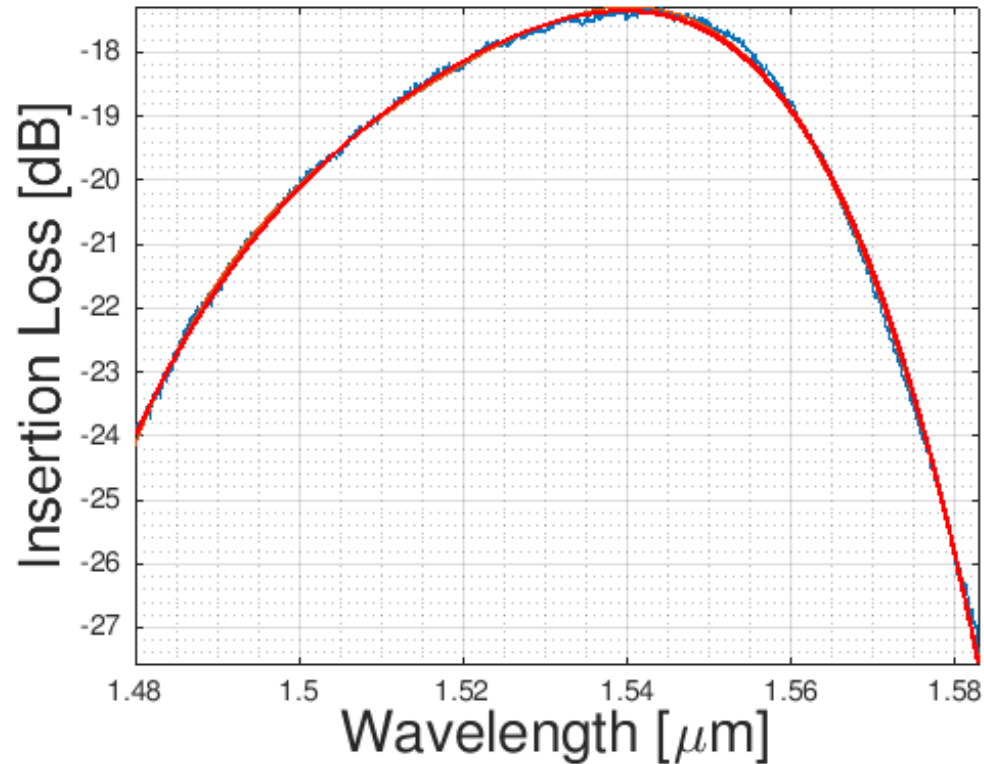
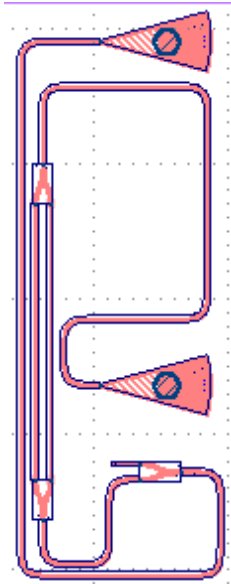
Simulation Data



Balanced MZI Using Y-Splitters (TM)

MZI Length Difference $\Delta L = 0$

Polarization: TM



Wavelength [μm]	TE Loss[dB]	TM Loss [dB]
1.52	35	18.2
1.55	17.9	17.9
1.58	26	27.5

Double-Bus Ring Resonators

Polarization: TE

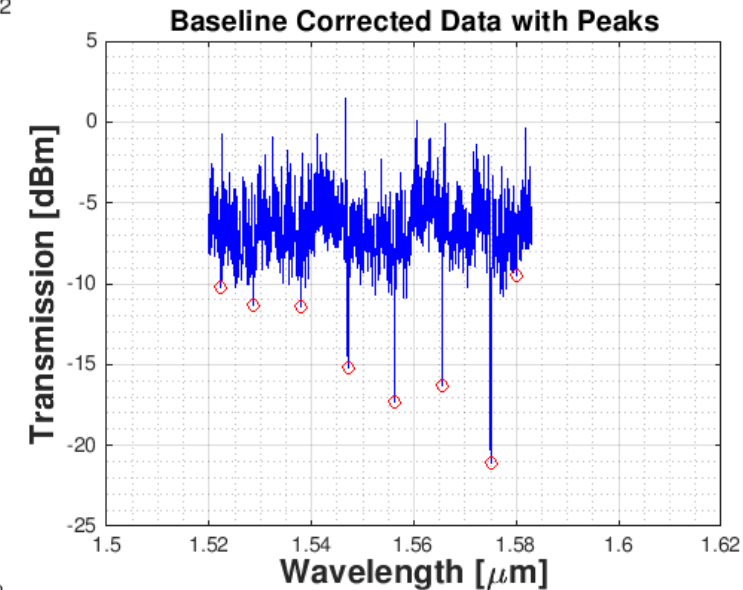
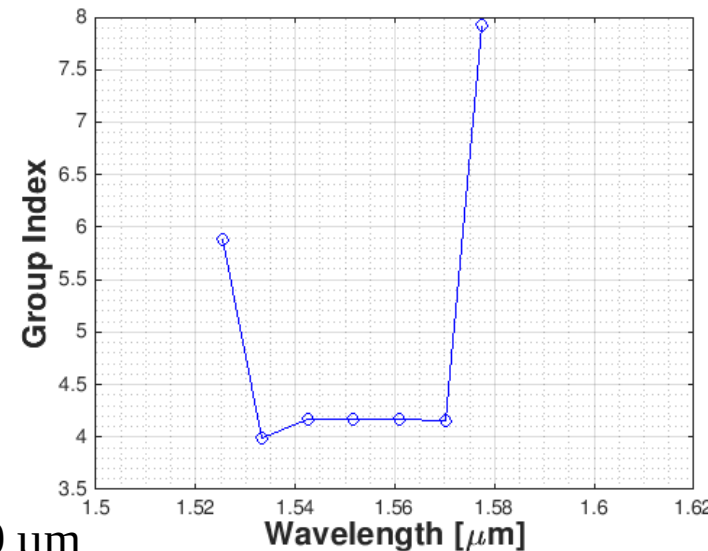
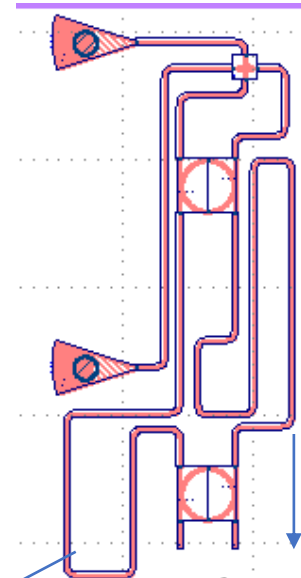
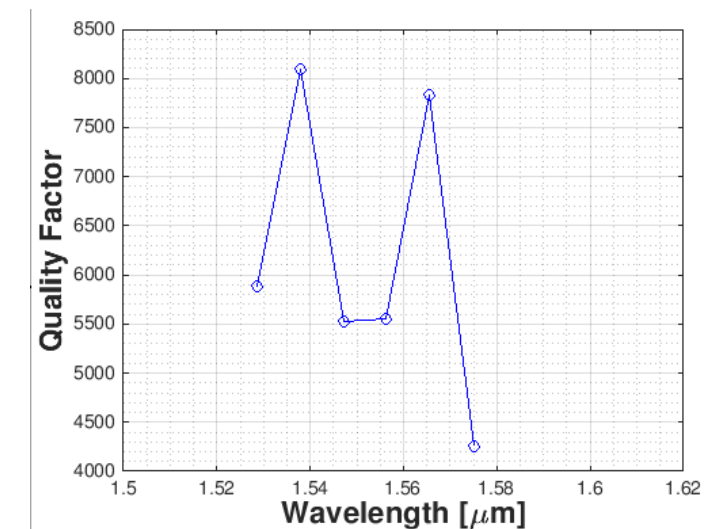
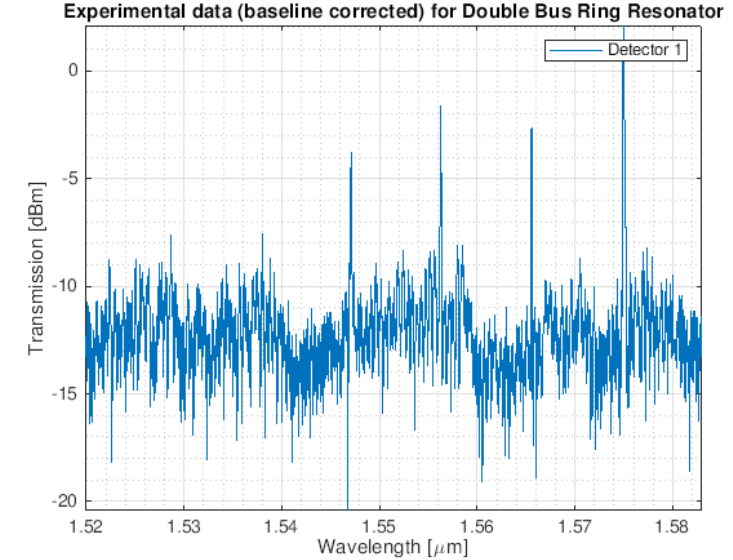
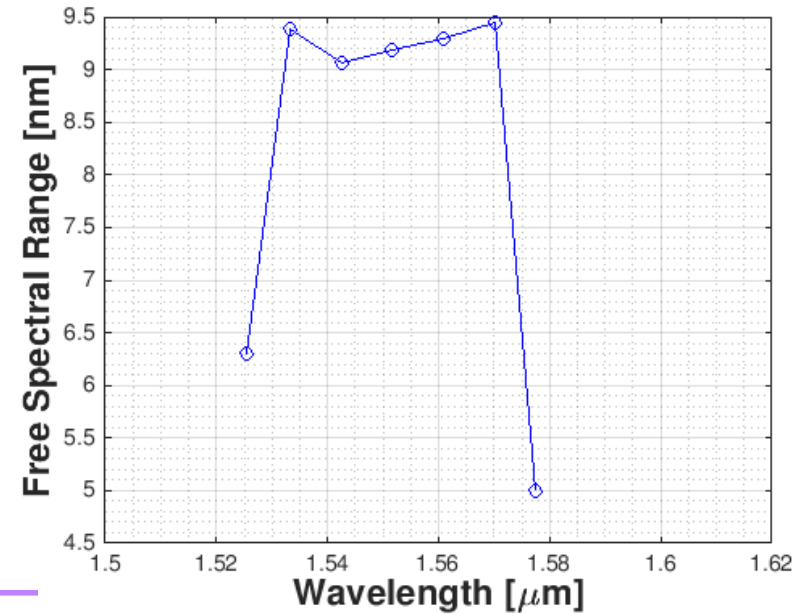
Following parameters are considered for finding peaks:

Minimum resonance separation: 9 nm

Minimum resonance depth: 4 dBm

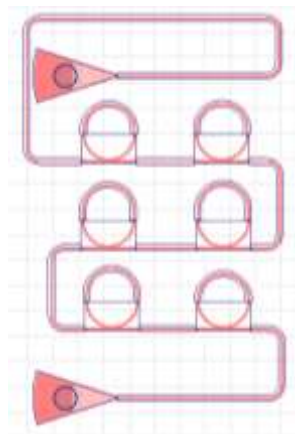
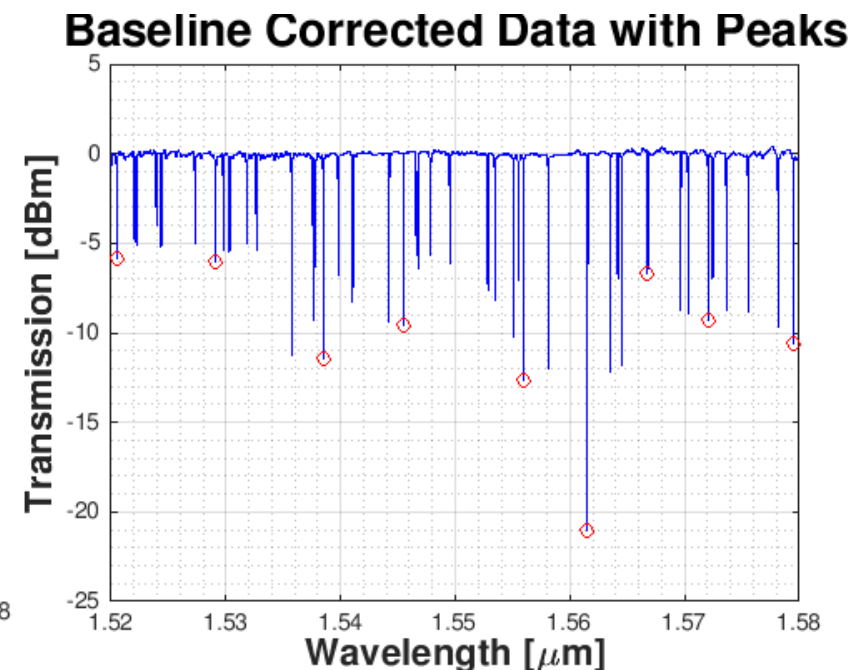
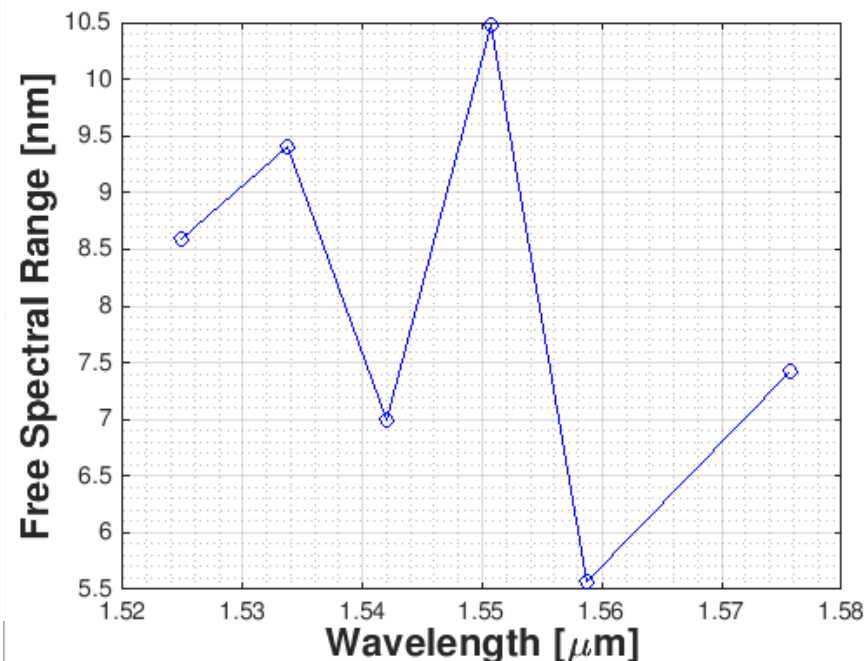
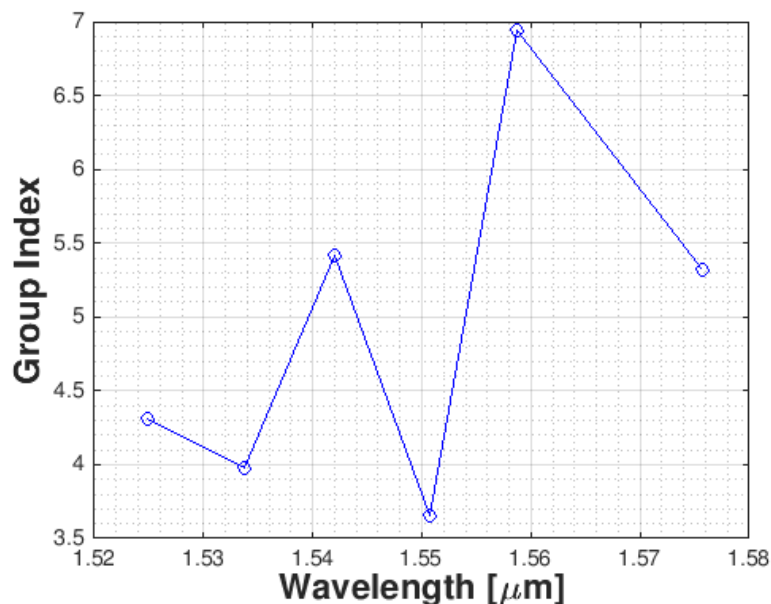
Transmission in Detector 1 (with grating couplers) <-60 dBm

The minima peaks, FSR, Quality Factor and Group Index values are derived using negative thru.



291.246 μm 357.89 μm

Loopback Using Six Single-Bus Ring Resonators



Polarization: TE
 Ring Gap: $0.2 \mu\text{m}$
 Ring Radius: $10 \mu\text{m}$
 Ring Circumference: $62.8 \mu\text{m}$
 Loopback length excluding rings: $531.68 \mu\text{m}$

To find the above peaks, the following parameters are considered:

Minimum resonance separation: 5 nm
 Minimum resonance depth: 4 dBm

Outline

1. E-Beam Data
2. **RIT Chip Data**

Loopback Using Single-Bus Ring Resonator: 250 nm Gap

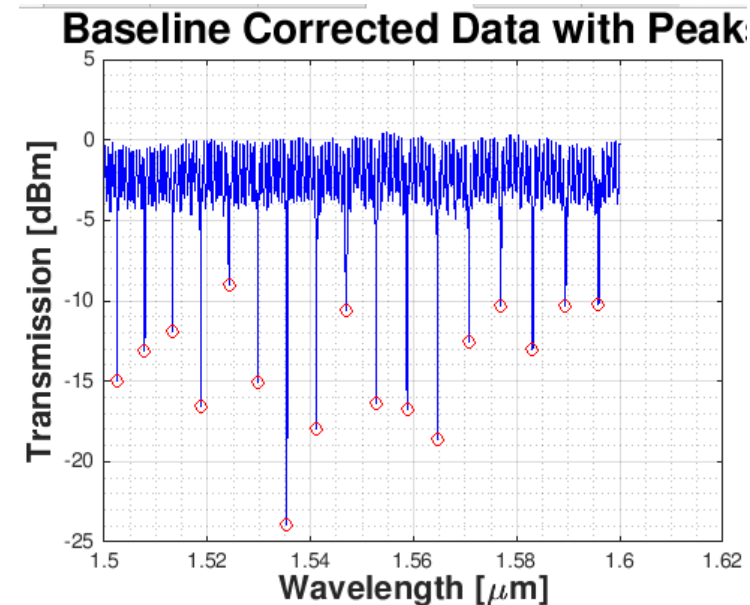
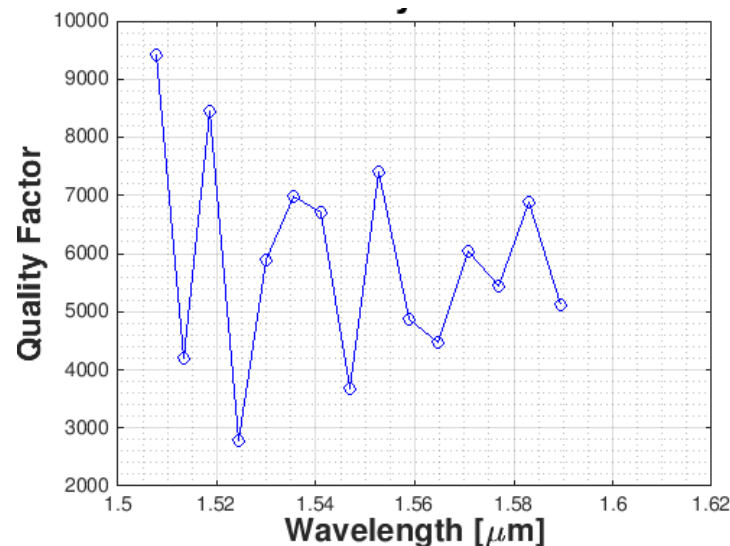
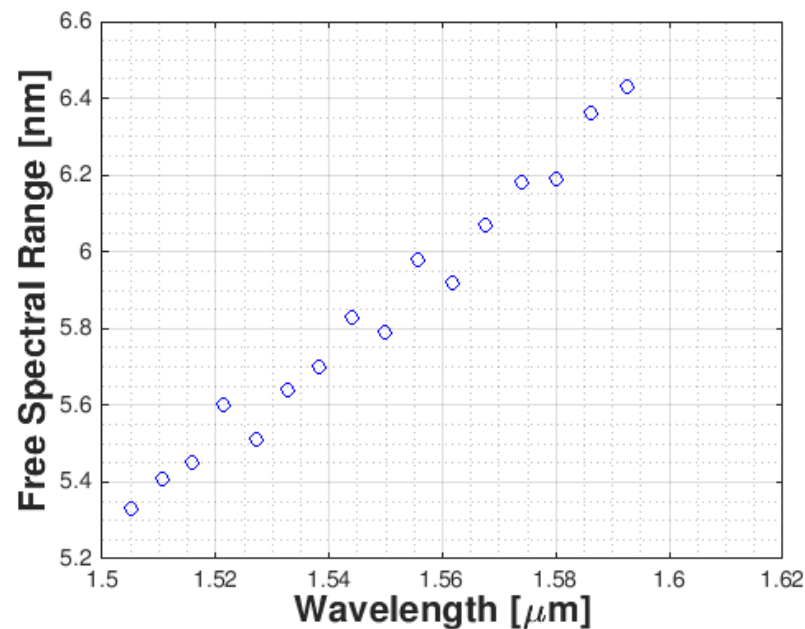
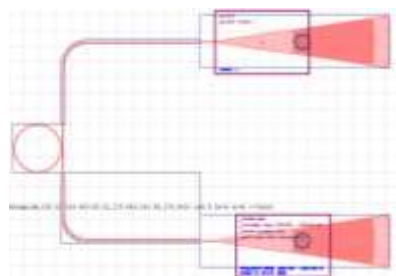
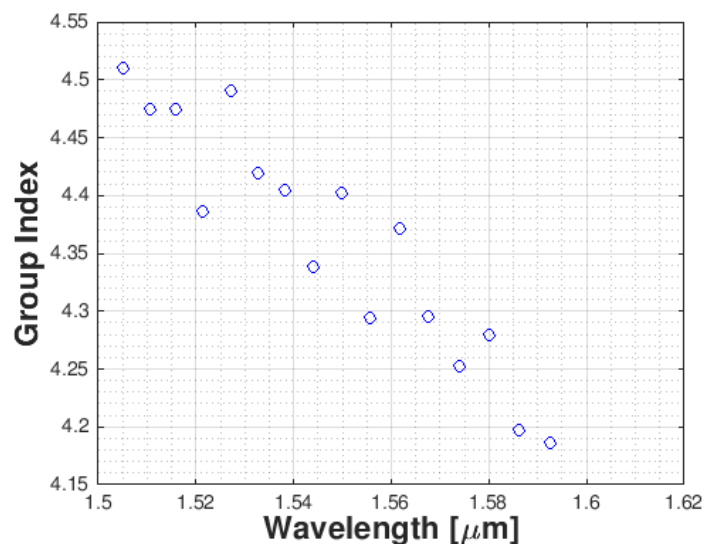
Ring Radius $R = 15\ \mu\text{m}$

Ring Gap = $0.25\ \mu\text{m}$

Ring Circumference = $2\pi R = 94.2\ \mu\text{m}$

Waveguide Length (excluding ring circumference) = $258.034\ \mu\text{m}$

Polarization: TM



250 nm can be considered a good gap since pretty good resonances are observed at this gap. However, as the gap is increased, data begins to mess up.

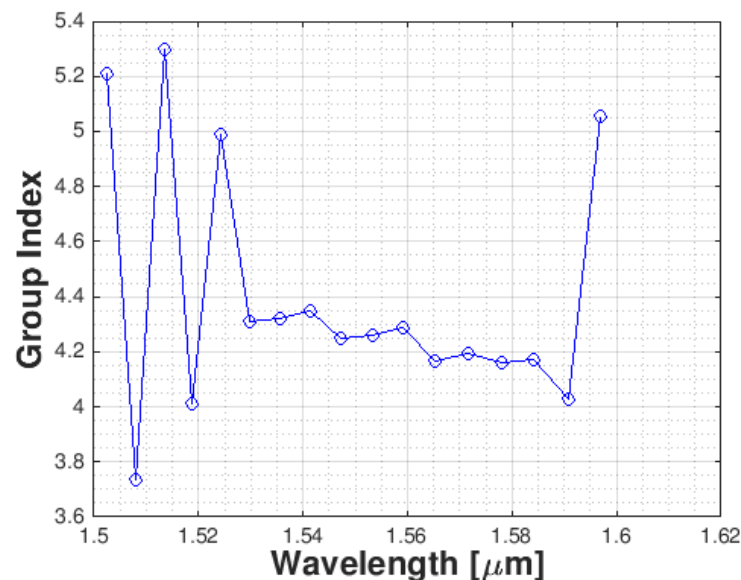
Loopback Using Single-Bus Ring Resonator: 400 nm Gap

Ring Radius $R = 15 \mu\text{m}$

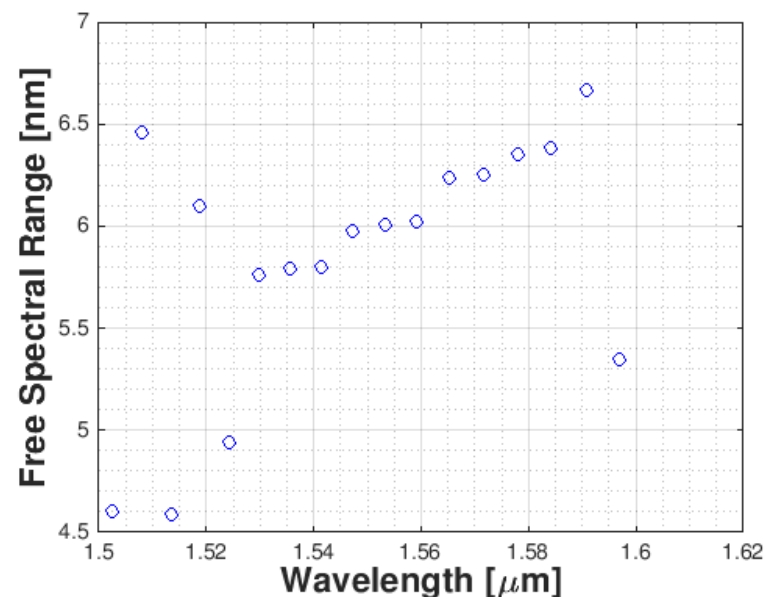
Ring Gap = $0.4 \mu\text{m}$

Ring Circumference = $2\pi R = 94.2 \mu\text{m}$

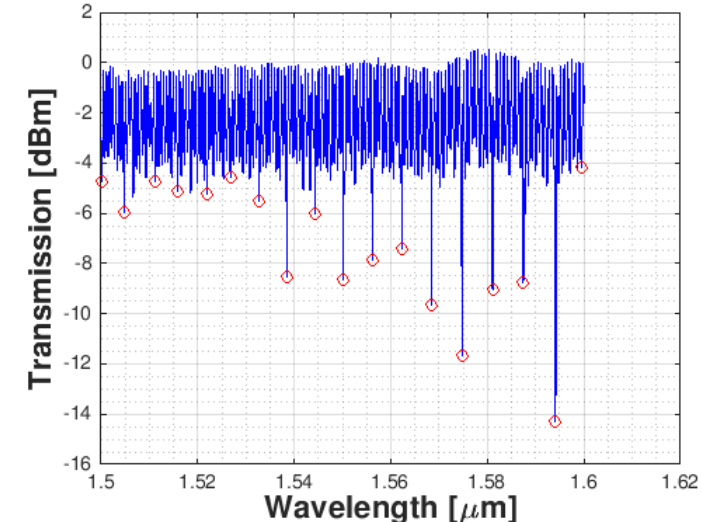
Waveguide Length (excluding ring circumference) = $258.034 \mu\text{m}$



Polarization: TM



Baseline Corrected Data with Peak

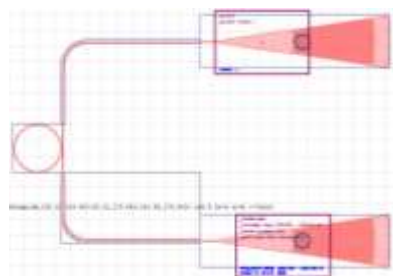


Quality of resonances begin to decline with bigger gaps.

To find minima peaks, following parameters are considered:

Minimum resonance separation: 5 nm

Minimum resonance depth: 4 dBm



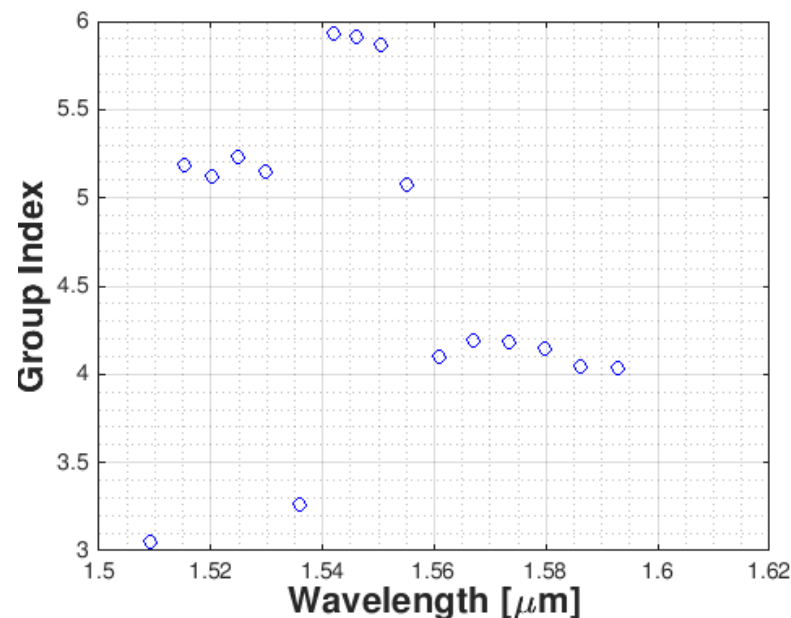
Loopback Using Single-Bus Ring Resonator : 500 nm Gap

Ring Radius $R = 15\text{ }\mu\text{m}$

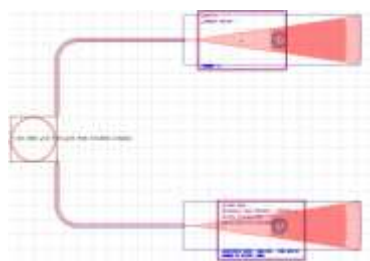
Ring Gap = $0.5\text{ }\mu\text{m}$

Ring Circumference = $2\pi R = 94.2\text{ }\mu\text{m}$

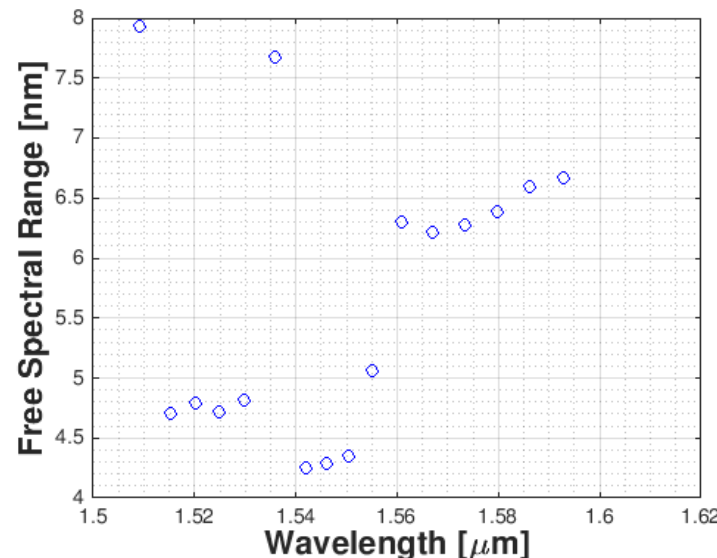
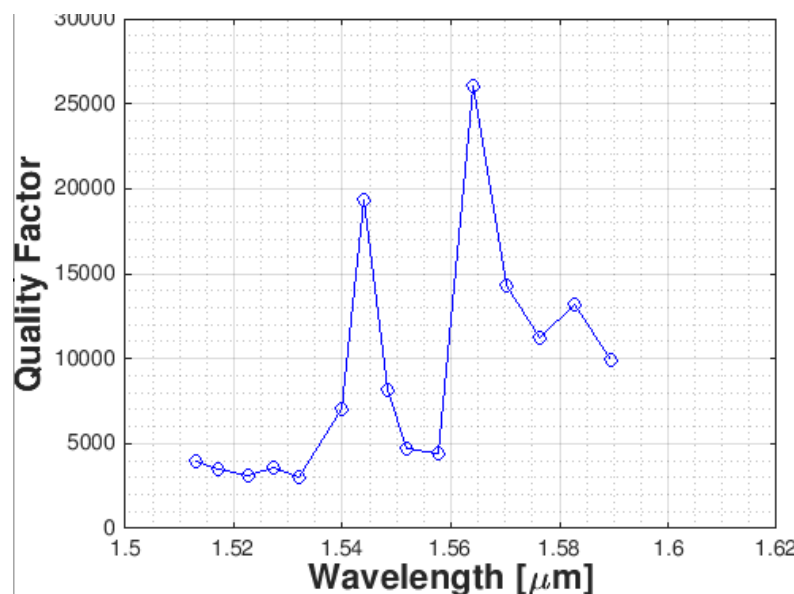
Waveguide Length (excluding ring circumference) = $258.034\text{ }\mu\text{m}$



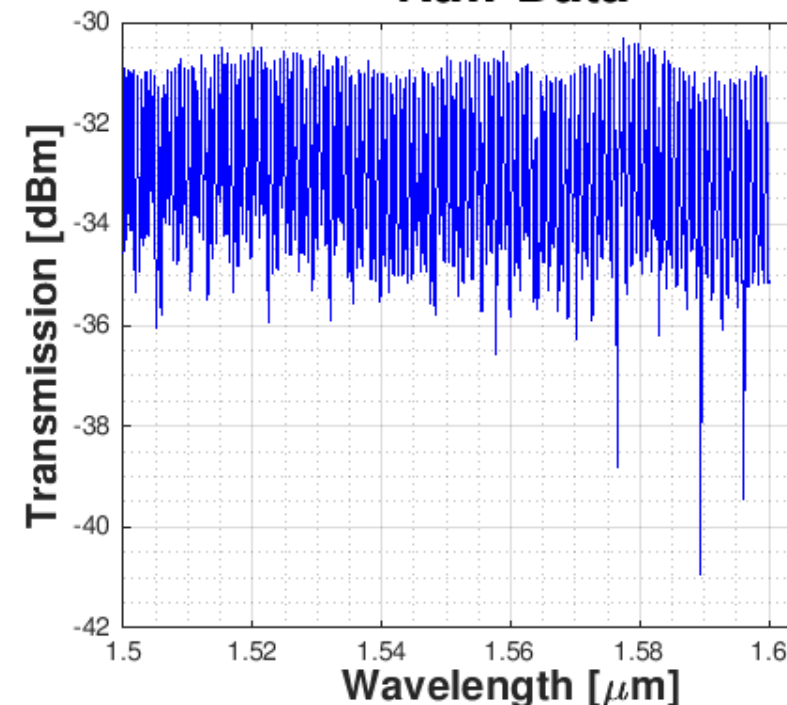
Group Index obtained is incredibly high



Polarization: TM



Raw Data



Due to high amount of reflections light is travelling back and forth due to which interference is happening, which is causing FSR to behave unexpectedly.

FSR @ $1.55\text{ }\mu\text{m} \approx 4.5\text{ nm}$

Highest quality factor (>25000) is obtained at $\lambda \approx 1.56\text{ }\mu\text{m}$

Loopback Using Single-Bus Ring Resonator : 700 nm Gap

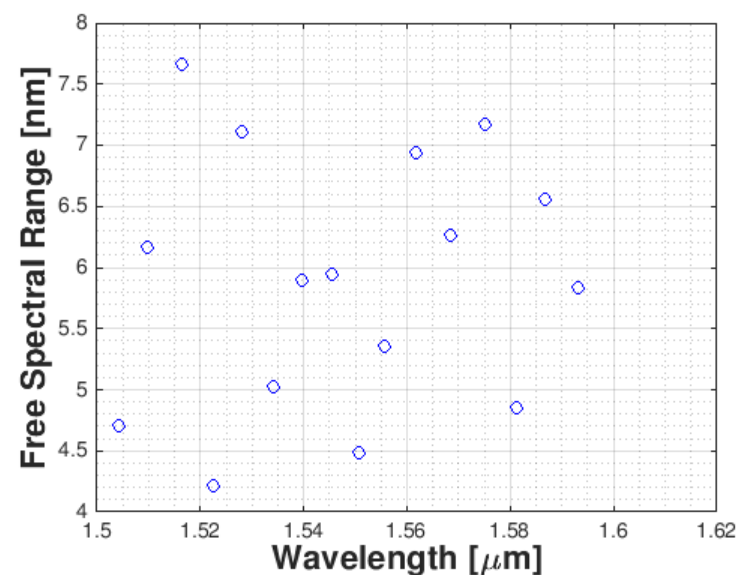
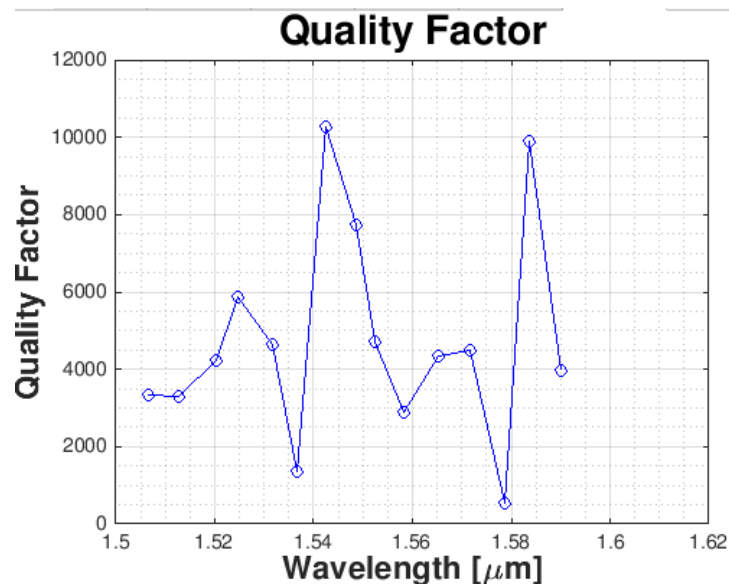
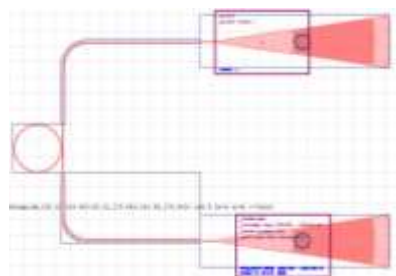
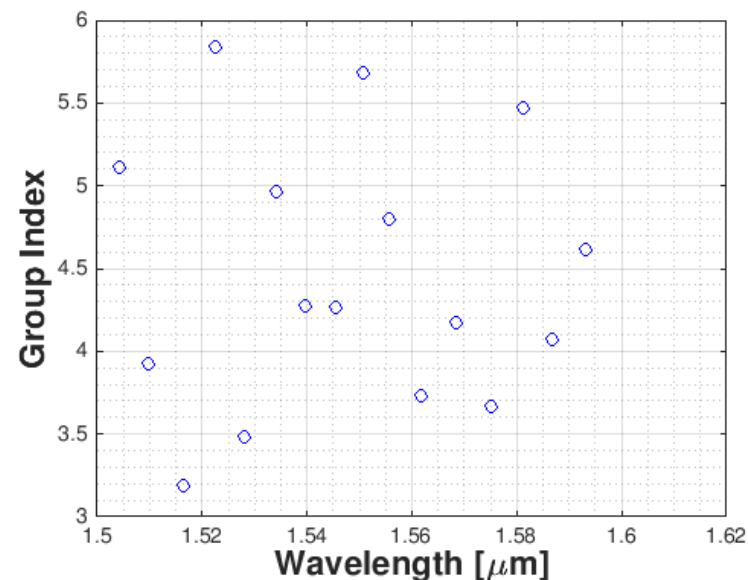
Ring Radius $R = 15 \mu\text{m}$

Ring Gap = $0.7 \mu\text{m}$

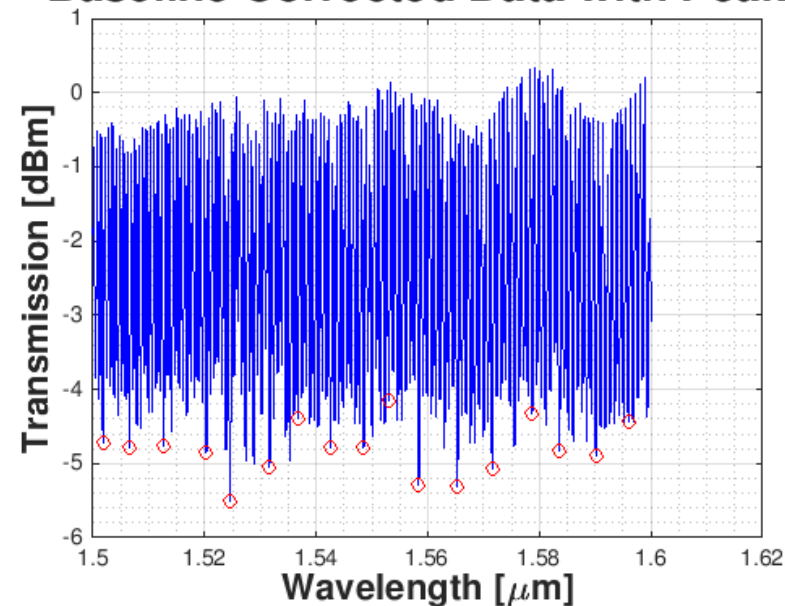
Ring Circumference = $2\pi R = 94.2 \mu\text{m}$

Waveguide Length (excluding ring circumference) = $258.034 \mu\text{m}$

Polarization: TM



Baseline Corrected Data with Peak

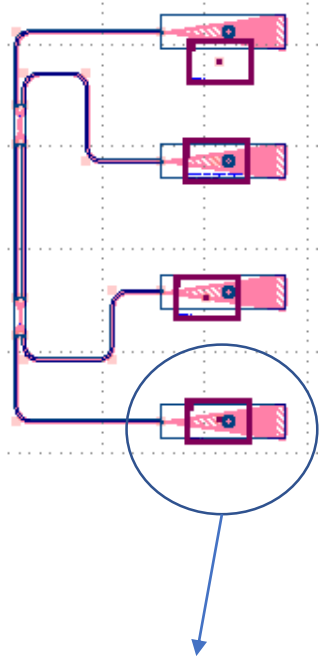


- Extremely noisy
- Very High amount of reflections
- Poor resonances

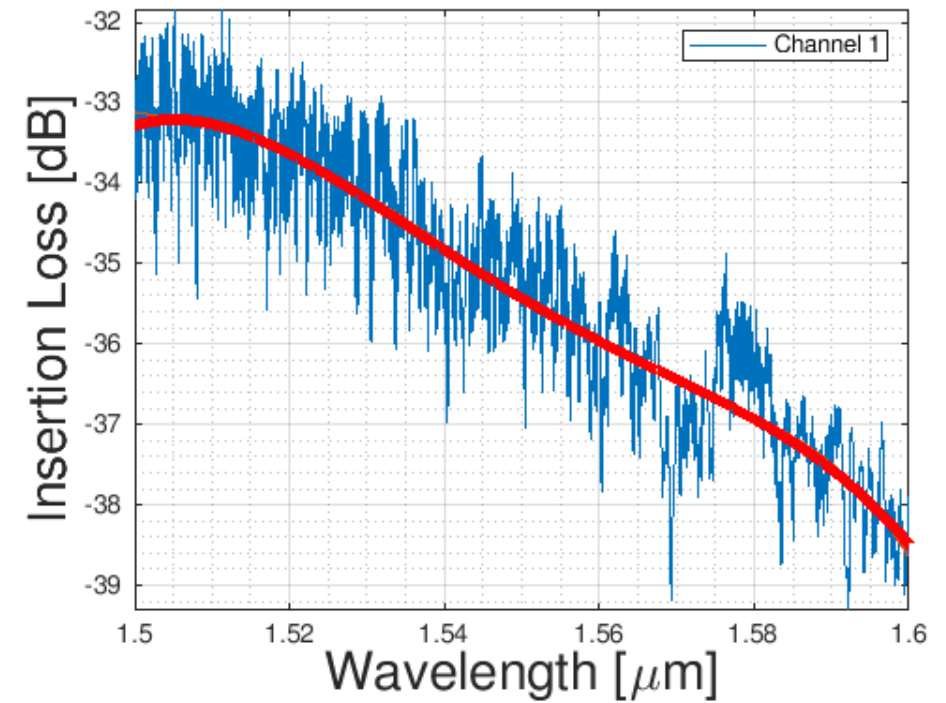
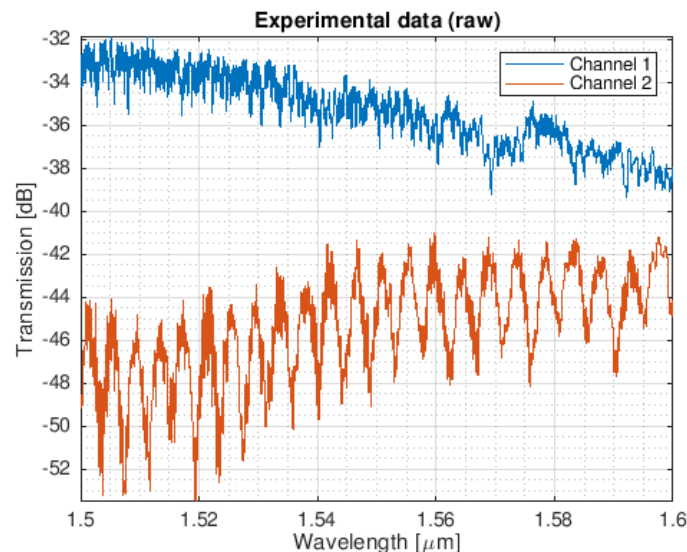
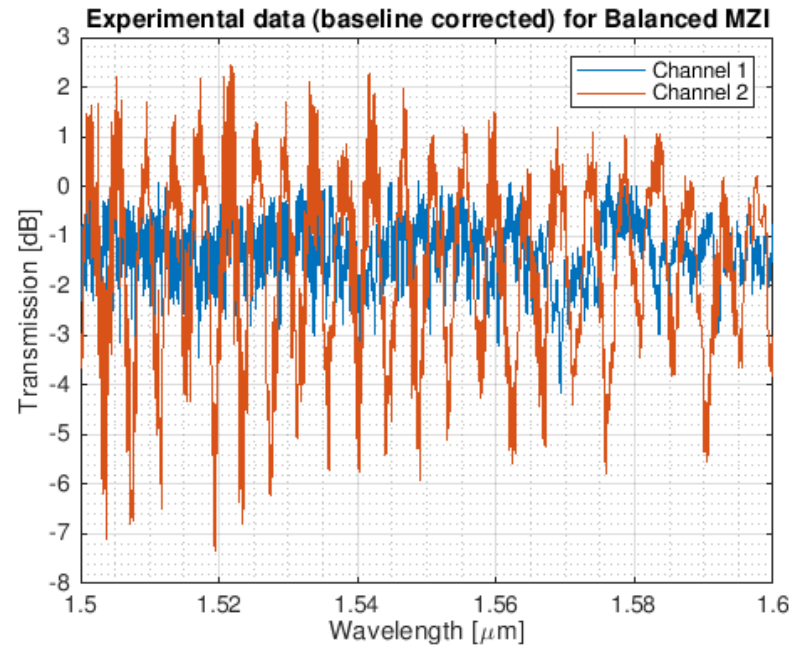
Balanced MZI Using Two 50:50 Broadband Directional Couplers

Length Difference $\Delta L = 0 \mu\text{m}$

Polarization: TM



Channel 1:
Expecting all power

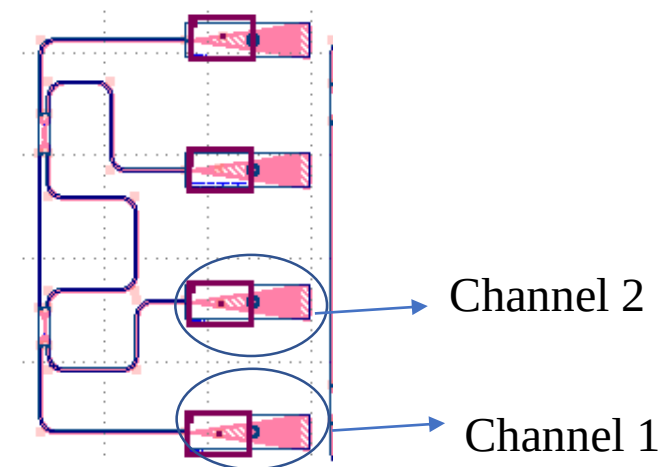
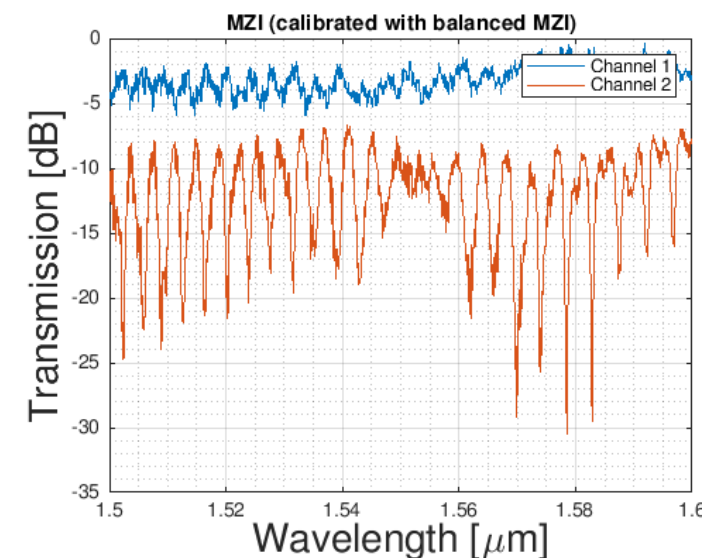
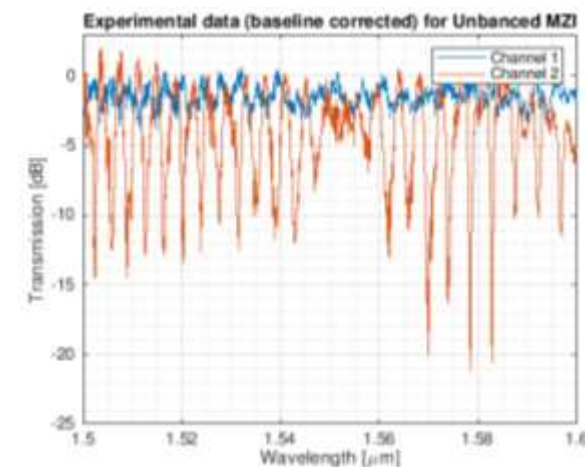
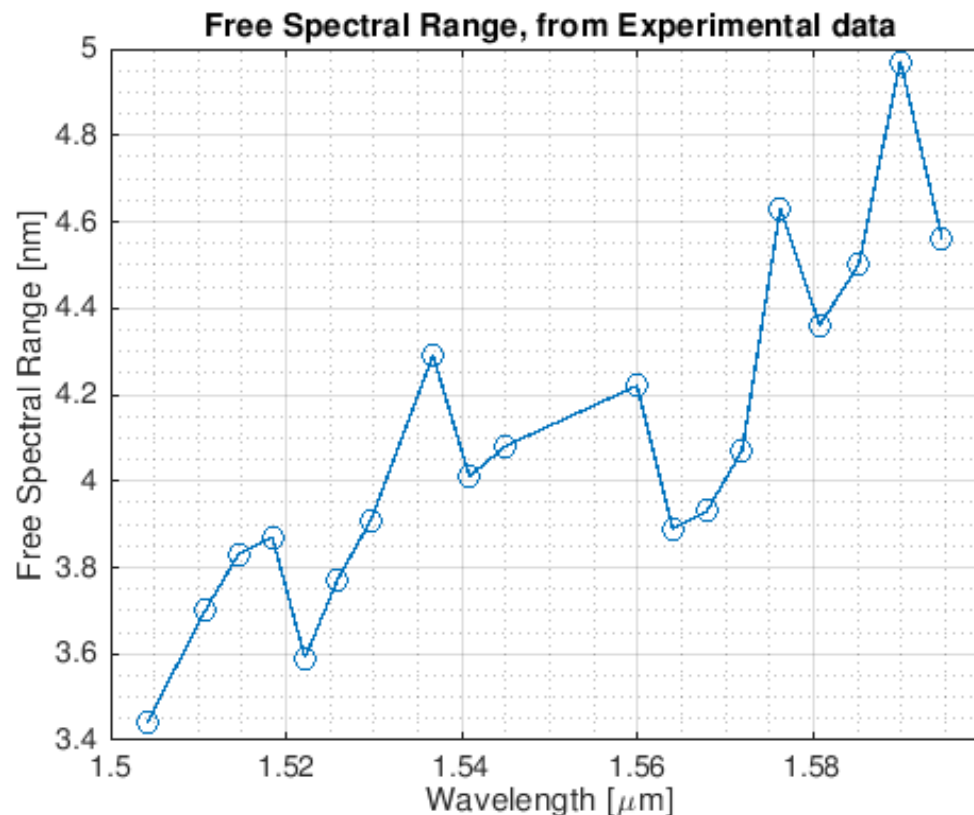
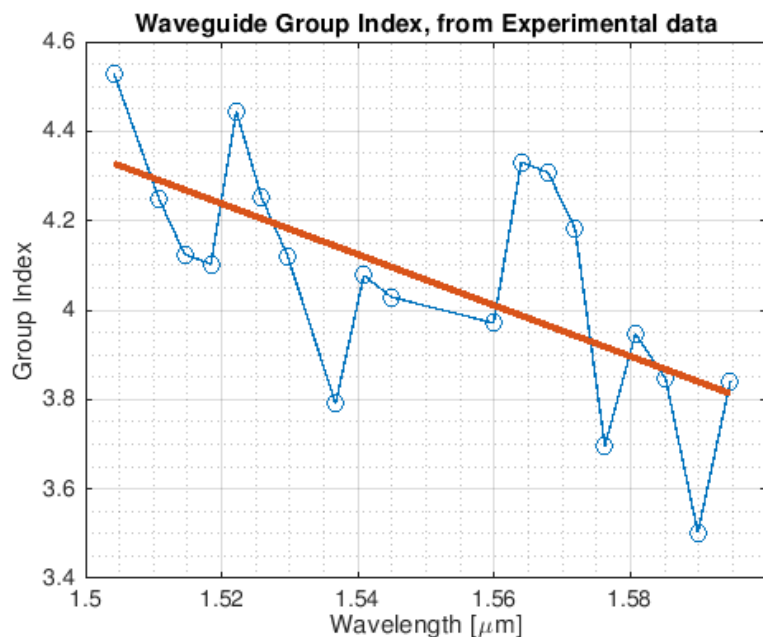


35 dB insertion loss observed at $1.55\mu\text{m}$

Unbalanced MZI Using Two 50:50 Broadband Directional Couplers

Length Difference $\Delta L = 145.19 \mu\text{m}$

Polarization: TM



Dispersion [ps/nm-km]: -18973.3205

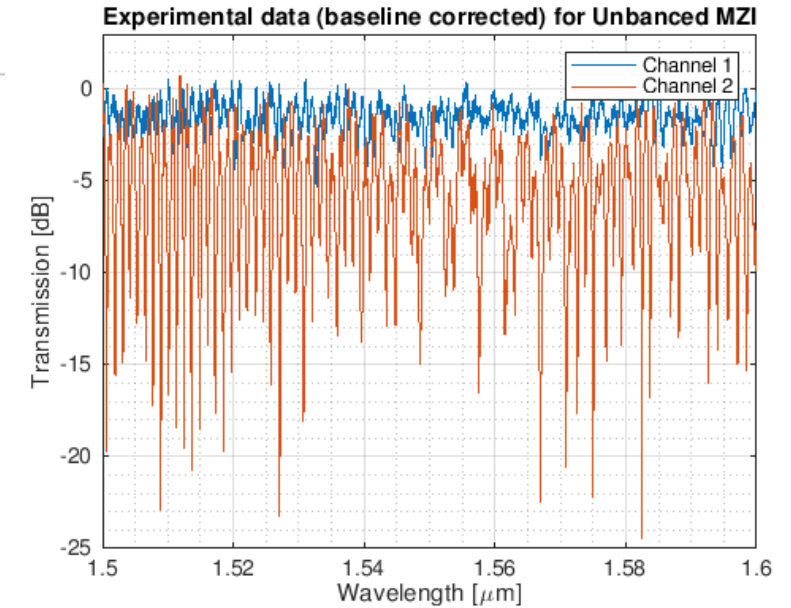
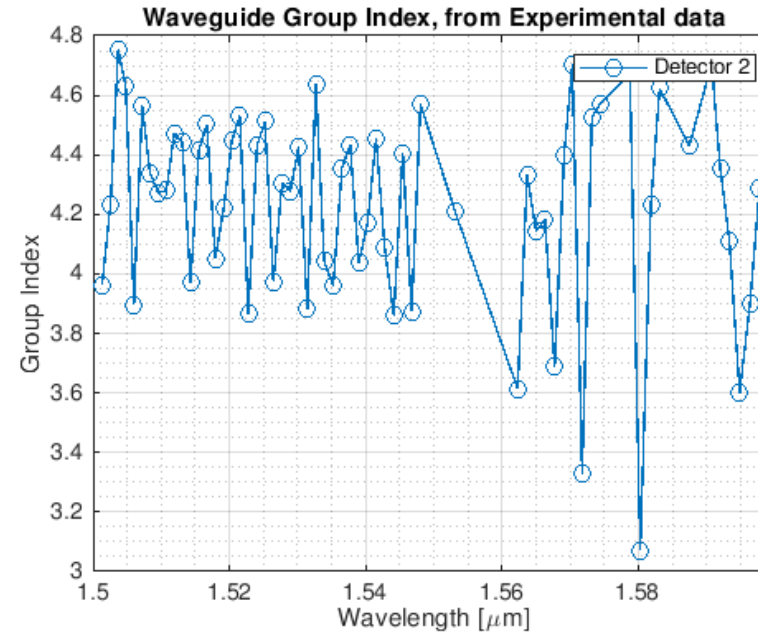
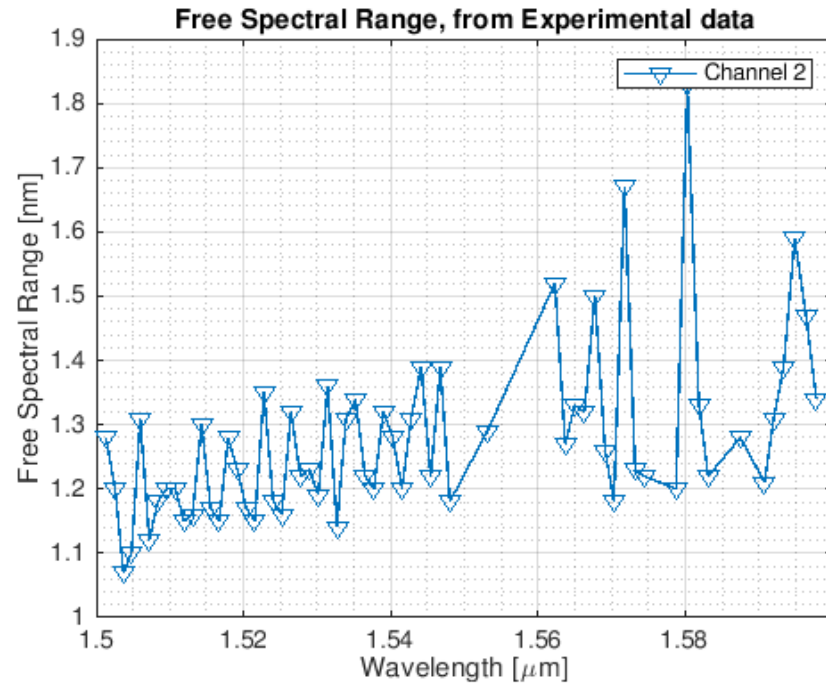
FSR $\approx 4.1 \text{ nm @ } 1.55 \mu\text{m}$

Fit parameters $(n_1, n_2, n_3) = (1.77, -1.6689, 1.6) \rightarrow 61.48\%$ good fit

Channel 1 is fairly balanced (no oscillations) hence did not get FSR or group index for it. The above FSR data and Group Index is obtained for channel 2.

Unbalanced MZI Using Two 50:50 Broadband Directional Couplers

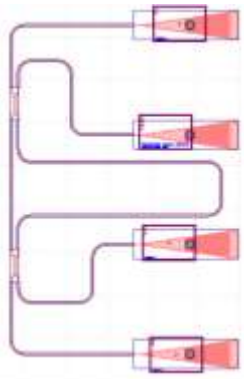
Length Difference $\Delta L = 444.326 \mu\text{m}$ Polarization: TM



The above FSR data and Group Index is obtained for channel 2.

Dispersion [ps/nm-km]: -6344.9947

FSR $\approx 1.2 \text{ nm}$ @ $1.55 \mu\text{m}$



Unbalanced MZI Using Two Y-Splitters and a Ring

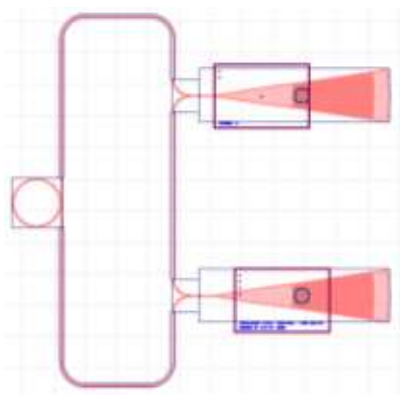
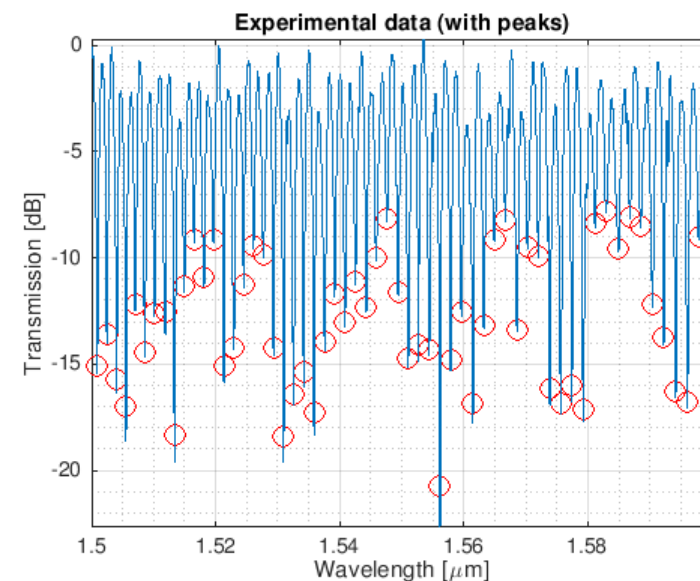
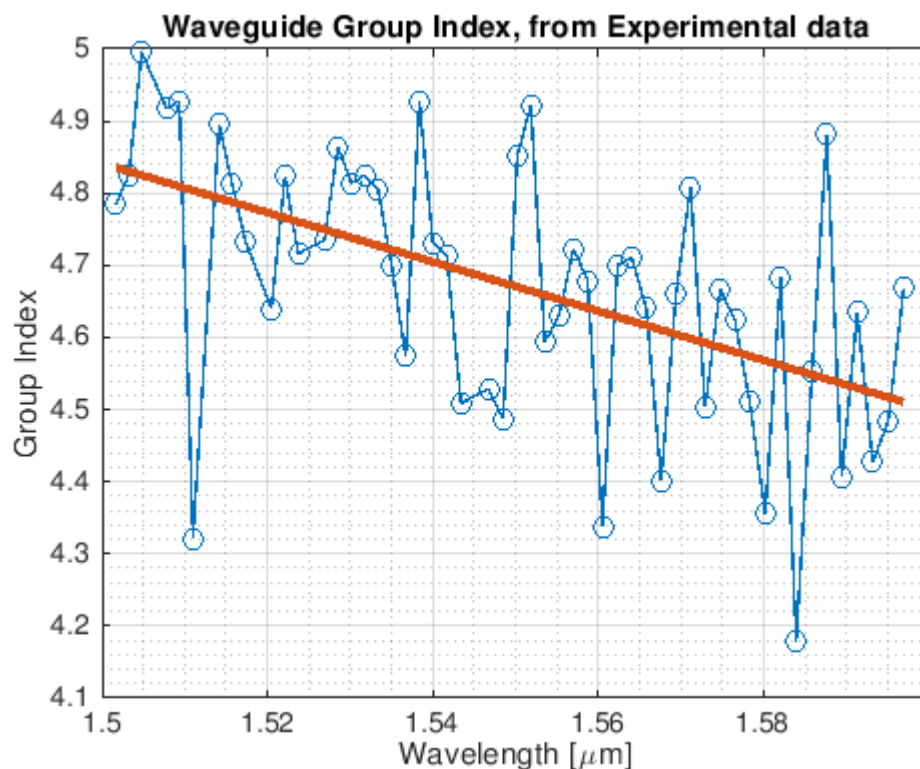
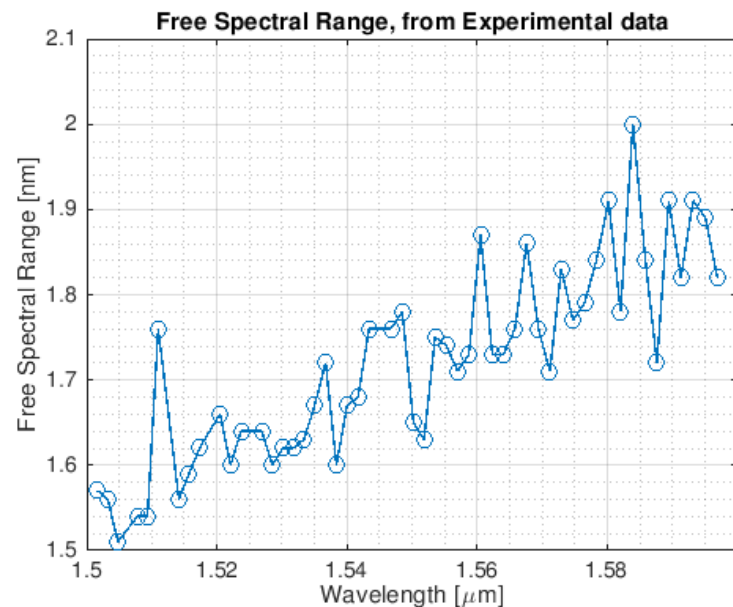
Ring Radius $R = 15 \mu\text{m}$

Ring Gap $= 0.4 \mu\text{m}$

Ring Circumference $= 2\pi R = 94.2 \mu\text{m}$

Length Difference $\Delta L = 300.168 \mu\text{m}$

Polarization: TM



Fit parameters (n_1, n_2, n_3): (1.769, -1.8767, 1.1035)

Goodness of fit, r^2 value: 0.81437

Waveguide parameters at wavelength [μm]: 1.5459

Group index: 4.7077

Dispersion [ps/nm-km]: -15476.6257

Obtained high value of group index for TM.

Unbalanced MZI Using Two Y-Splitters and a Ring

Polarization: TM

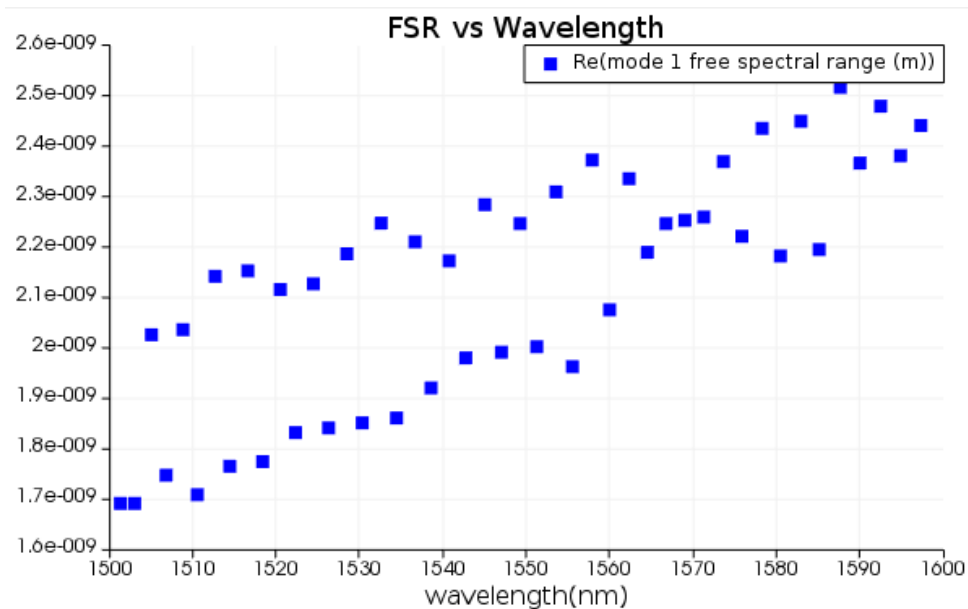
Ring Radius $R = 15 \mu\text{m}$

Ring Gap $= 0.25 \mu\text{m}$

Ring Circumference $= 2\pi R = 94.2 \mu\text{m}$

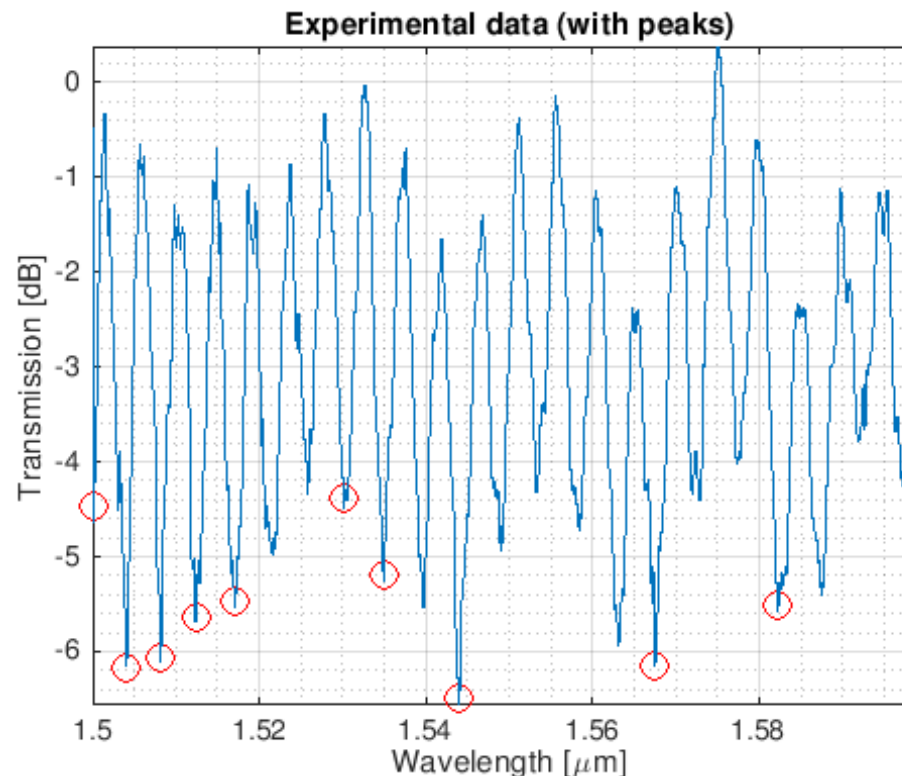
Length Difference $\Delta L = 300.168 \mu\text{m}$

Following FSR data is obtained using simulation result:



$\text{FSR}@ \lambda = 1.55 \mu\text{m} \approx 2.2 \text{ nm}$

$n_g = (1.55^2 \times 1000) / (2.2 \times 300.168) = 3.64$



Group Index values could not be found using MATLAB script.

